

Layered Double Hydroxides for CO₂ Capture: Structure Evolution and Regeneration

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Mg–Al–CO₃ layered double hydroxide (LDH) was synthesized, and its thermal evolution was investigated using X-ray diffraction, FTIR spectroscopy, and thermogravimetric analysis (TGA). The resultant LDH derivatives showed excellent CO₂ adsorption capabilities, especially suitable for high-temperature CO₂ separation from flue gases. Calcination of crystalline LDHs at 400 °C led to phase transformation yielding amorphous Mg–Al mixed oxides having a CO₂ sorption capacity of 0.49 mmol/g at 200 °C. Reversible and irreversible CO₂ sorption was determined to be ~88% and ~12% of the total CO₂ sorption, respectively. Regeneration restored the Mg–Al mixed oxide to 98% of its initial CO₂ sorption after several cycles of CO₂ adsorption testing. This clearly indicates the desirable properties of Mg–Al mixed oxide for CO₂ capture from flue gases at high temperatures (up to 200 °C).

Introduction

Atmospheric concentrations of several green house gases, in particular, carbon dioxide (CO₂), have increased drastically in the recent past, largely as a result of the combustion of fossil fuels.¹ The capture of CO₂ from industrial flue gases has therefore become a global issue attracting world attention. Several options have been proposed to control CO₂ emissions, including increasing the efficiency of fossil-fuel combustion systems or finding suitable alternatives with renewable energy sources. Another viable option to control CO₂ emissions is carbon dioxide sequestration, that is, capturing and securely storing CO₂ emitted from major sources of emission. The existing methods available for CO₂ sequestration include absorption by physical and chemical wet scrubbing, adsorption by solids using pressure and temperature swing modes, cryogenic distillation, CO₂-selective membranes, and mineralization processes.^{2,3} However, each of these systems has its own limitations that impede their technical or economic viability in CO₂ postcombustion capture systems.

Selective adsorption is another promising technique considered for CO₂ separation. The most widely used gas separation technologies, such as pressure swing adsorption (PSA) and temperature swing adsorption (TSA), have been applied for CO₂ separation mainly in petrochemical processes.^{5–10} A few inorganic materials (zeolites and activated carbons) have been found to have good adsorption capacities for CO₂. However, they are not attractive for high-temperature applications because the CO₂ adsorption capacities of these materials decrease drastically with temperature.^{4,11} Bearing in mind the limitations of the existing methods and materials, there is an intense search for new adsorbents that exhibit high selectivities and good adsorption capacities for CO₂ at high temperature. In addition, adequate adsorption/desorption kinetics, stable adsorption capacity after repeated adsorption/desorption cycles, and good mechanical strength of the adsorbent material after cyclic exposure are all desirable for a good CO₂ adsorbent.^{12,13} Layered double hydroxides (LDHs) are an interesting class of inorganic

compounds, and in particular, their derivatives produced upon calcination have desired properties as CO₂ adsorbents in postcombustion capture applications.

LDHs belong to a large group of anionic or basic clays that are often referred as hydrotalcite-like compounds (HTlcs).¹⁴ These materials have received much attention because of their wide range of applications as catalysts, precursors, and adsorbents.^{15–17} Their structure consists of positively charged brucite-like layers of the form $[M^{2+}_{1-x}M^{3+}_x(OH)_2]^{x+}$ with trivalent cations partially substituting for divalent cations. The excess positive charges attained because of the trivalent cations in the hydroxide layers are compensated by anions as well as water molecules in the interlayer regions, i.e., $[A^{m-}_{x/m} \cdot nH_2O]^{x-}$. LDHs are thus represented by a general formula $[M^{2+}_{1-x}M^{3+}_x(OH)_2]^{x+} \cdot [A^{m-}_{x/m} \cdot nH_2O]^{x-}$, where M^{2+} is a divalent cation such as Mg²⁺, Ni²⁺, Zn²⁺, Cu²⁺, or Mn²⁺; M^{3+} is a trivalent cation such as Al³⁺, Fe³⁺, or Cr³⁺; and A^{m-} is an interlayer anion such as CO₃²⁻, SO₄²⁻, NO₃⁻, Cl⁻, or OH⁻. The value of x is typically between 0.17 and 0.33, even though there is no strict limitation to this value. A typical structural representation of an LDH is shown in Figure 1.¹⁴

Recent reports in the literature suggest the potential of CO₂ adsorption using LDH derivatives produced upon calcination.^{19–28} In one typical CO₂ adsorption study conducted at higher temperatures, Yong et al. found an adsorption capacity of 0.5 mol/kg at 300 °C and 1 bar.^{22,23} The thermal evolution of the Mg–Al–CO₃ LDH structure is considered to be crucial in determining the CO₂ adsorption capacity, and this issue has been extensively investigated in the recent studies.^{4,18–21} These investigations revealed that LDHs undergo interlayer water dehydration, dehydroxylation of layered OH groups, and release of interlayer CO₃²⁻ groups in various temperature regimes, finally leading to the formation of an amorphous Mg/Al mixed solid oxide with a larger surface area and good stability at high temperatures, which makes the mixed oxide a viable material for CO₂ sorption. In this particular study, we investigated the thermal evolution pattern of the LDH structure and CO₂ sorption, as well as the regeneration properties of the resultant mixed oxide (hereafter referred to as LDO). The structural evolution of the mixed oxide (LDO) during postcalcination storage/aging in air and its negative influence on the CO₂ sorption capacity

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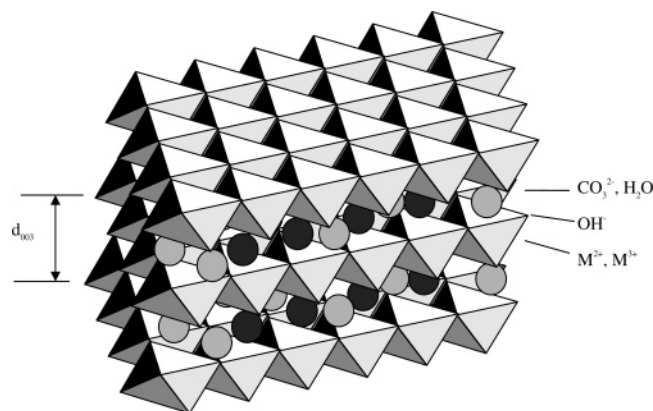


Figure 1. Schematic structural representation of Mg–Al–CO₃ LDH.

were studied using a wide range of techniques. Alternative methods such as vacuum storage and in situ calcination are suggested to overcome these effects.

Experimental Section

Material Preparation. The Mg–Al–CO₃ LDHs used in this study were prepared by coprecipitation of Mg²⁺ and Al³⁺ ions in an alkaline solution containing NaOH/Na₂CO₃. Specifically, 1.0 M solutions of Mg(NO₃)₂·6H₂O and Al(NO₃)₃·9H₂O were mixed in a 3:1 ratio, and 150 mL of the mixed solution was added dropwise to 200 mL of a base solution containing 1.6 M NaOH and 0.1 M Na₂CO₃. The resultant mixture was refluxed at 80 °C for 24 h with continuous stirring. The LDH was separated by high-speed centrifugation and washed 5–6 times with deionized water to remove water-soluble sodium salts. Elemental analysis also confirmed that the sodium content was negligible. The LDH sample was dried at 100 °C in an oven and is referred to as raw LDH. This raw LDH had a nominal structural formula of Mg₃Al(OH)₈(CO₃)_{0.5}·2H₂O, as the Mg/Al atomic ratio determined by inductively coupled plasma–atomic emission spectroscopy (ICP–AES) on a Varian Vista Pro instrument was 2.9. Several portions of the raw LDH were calcined at different temperatures ranging from 200 to 600 °C to determine the optimum calcination temperature for achieving the optimal CO₂ sorption capacity. Calcination was carried out at a pressure of 5 kPa for 4 h. From the initial sorption screening results, 400 °C was found to be the optimum calcination temperature for all remaining experiments in the study. The LDH calcined at 400 °C is referred to as fresh LDO. A portion of the fresh LDO was stored for 48 h in an open sample bottle under atmospheric conditions and is referred to as aged LDO.

Structural Characteristics. FTIR spectroscopy, X-ray diffraction (XRD), and thermogravimetric analysis (TGA) were used to identify the LDH structure. In addition, these techniques were also used to monitor the evolution of the structure during the calcination and postcalcination (aging) periods. A Perkin-Elmer FTIR spectrometer (Spectrum 2000) was used to record the infrared spectra, and the XRD patterns were recorded using a Rigaku Miniflex X-ray diffractometer (with a variable slit width). A Shimadzu TG50 TGA instrument was used to investigate the thermal decomposition patterns of the raw LDH and the fresh and aged LDO samples. All thermogravimetric analysis experiments were conducted in inert helium (He) atmosphere. The surface area (BET) and pore size distribution were calculated by nitrogen adsorption at –196 °C using a Quantachrome Autosorb instrument. The isotherms were measured using LDH samples calcined at different temperatures under vacuum for at least 4 h.

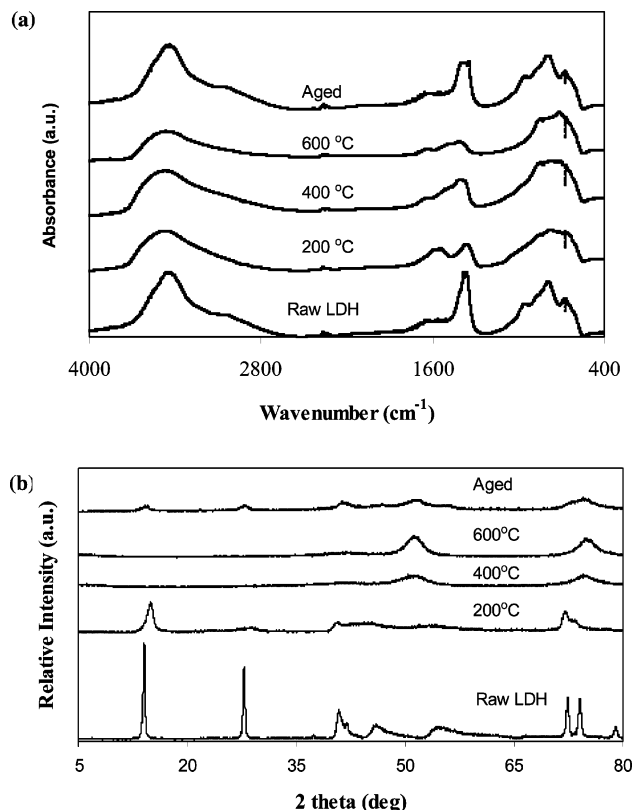


Figure 2. (a) FTIR spectra and (b) XRD patterns of raw LDH, LDHs calcined at different temperatures, and aged LDO.

CO₂ Sorption. High-temperature CO₂ sorption measurements were carried out using a Quantachrome Autosorb instrument with a chemisorption furnace. Both the fresh and aged LDO samples used for CO₂ sorption were degassed at 200 °C for 4 h and subsequently reweighed before measurements. Initially, sorption measurements were carried out at temperatures of 100, 200, 300, and 400 °C to determine the optimal sorption conditions for high-temperature CO₂-separation applications from flue gases. All remaining sorption measurements were conducted at 200 °C as the LDOs showed the highest CO₂ sorption at 200 °C under the current conditions. Reversible and irreversible sorption components were determined in the following manner: The first CO₂ sorption measured for an LDO sample was recorded as the combined or total sorption. The CO₂-adsorbed sample was then evacuated in situ at a pressure of 5 kPa at same temperature (200 °C) for 4 h and then subjected to sorption measurements similar to the first measurement. The sorption value obtained this time is denoted as reversible or weak sorption. The irreversible or strong sorption was then calculated indirectly by subtracting the reversible part from the combined or total sorption.⁴ Regeneration of LDOs was accomplished by recalcining the samples under conditions similar to those used for the initial calcination, e.g., 400 °C for 4 h at 5 kPa.

Results and Discussion

Structural Aspects. As shown in Figure 2, the transformation of the crystalline LDH phase to nearly amorphous mixed oxide was characterized using XRD and FTIR spectroscopy. In the XRD patterns of raw LDH (Figure 2b), the basal reflections from the (003), (006), etc., planes and the reflection from the (110) plane are indicative of the formation of a Mg–Al–CO₃ LDH material with an interlayer spacing of 0.760 nm and a

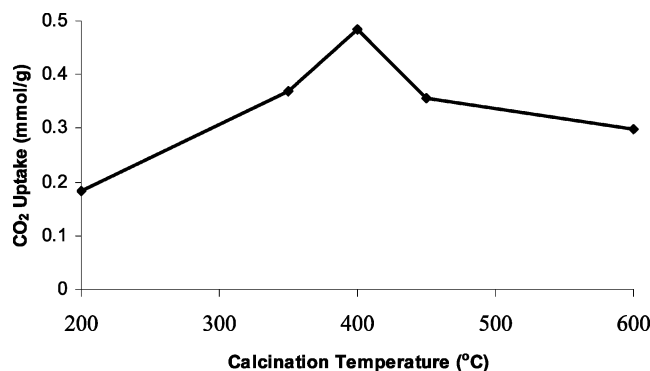


Figure 3. CO₂ sorption as a function of calcination temperature.

unit cell parameter a of 0.305 nm. The presence of carbonate in the interlayer is indicated by its characteristic vibration at 1360 cm^{-1} in the FTIR spectrum (Figure 2a). Further, the broad band at 3400 cm^{-1} with a shoulder at 3100 cm^{-1} and another broad band at 780 cm^{-1} are attributed to the O–H and M–O bond stretching vibrations in the raw LDH phase, respectively. Upon calcination, the crystalline LDH phase progressively changes to amorphous Mg–Al mixed oxide with increasing calcination temperature. The XRD patterns in Figure 2b indicate that the LDH layered structure partially collapses after calcination at 200 °C and then completely disappears at 400 °C. The removal of CO₂ continues to calcination temperatures over 600 °C, as the weak IR peak at 1360 cm^{-1} can still be observed (Figure 2a). The resultant LDOs derived at 400 and 600 °C have two broad peaks at 52° and 75° in the XRD patterns (Figure 2b), which corresponds to the periclase MgO phase (JCPDS 43-1022) being nearly amorphous.

CO₂ Sorption. Figure 3 shows the CO₂ sorption capacities of LDH derivatives measured at 200 °C as a function of LDH calcination temperature. The profile indicates that the sample calcined at 400 °C has a maximum sorption capacity, which is presumably due to the tradeoff between the surface area and availability of active basic sites. The BET surface area values measured for these samples clearly support the observed behavior. The sample calcined at 400 °C, which is basically an amorphous mixed oxide phase, was found to have the maximum BET surface area of 167 $\text{m}^2\cdot\text{g}^{-1}$ compared to the samples calcined at 200 and 600 °C. The sample calcined at 200 °C is considered to be a dehydrated LDH, still keeping its layered structure intact with the CO_3^{2-} ions occupying the basic sites. Hence, this sample has a relatively small BET surface area (63 $\text{m}^2\cdot\text{g}^{-1}$) and fewer basic sites, which leads to a low CO₂ sorption capacity. On the other hand, calcination of the LDH at higher calcination temperatures (500–900 °C), e.g., at 600 °C, can decompose most of the CO_3^{2-} to release some basic sites for CO₂ sorption, but results in a low surface area (82 $\text{m}^2\cdot\text{g}^{-1}$ at 600 °C) because of the phase transformation of amorphous mixed oxide into crystalline spinel and MgO phases.^{4,18–21} As a result, the CO₂ sorption cannot reach the maximum value. Calcination at 400 °C not only produced the largest surface area, but also generated enough basic sites by decomposing CO_3^{2-} , giving rise to the highest CO₂ sorption amount.

Further, sorption capacities were also measured at 100, 200, 300, and 400 °C using the sample calcined at 400 °C. The results presented in Table 1 show that the sorption capacity reached its maximum at 200 °C. Even though the sorption values show a general trend of decreasing with increasing temperature, the value observed at 100 °C shows a deviation with a lower sorption value. It is well understood that CO₂ sorption on an LDO is a combined value resulting from both physical and

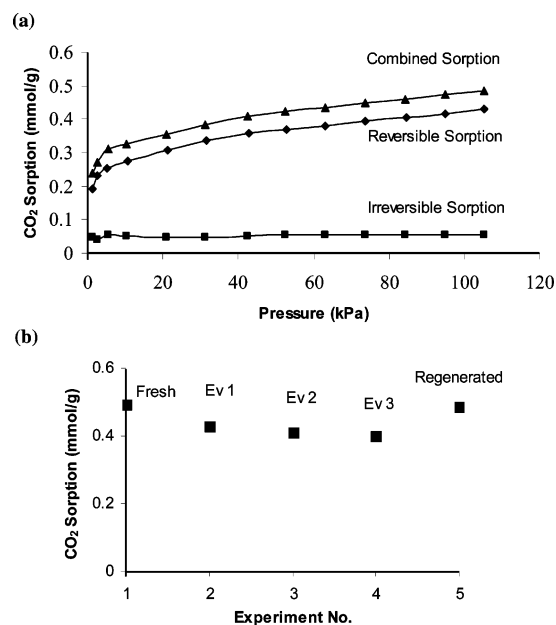


Figure 4. (a) Combined, reversible, and irreversible CO₂ sorption on LDO. (b) Sorption values after three vacuum evacuations (Ev1, Ev2, Ev3) and regeneration.

Table 1. CO₂ Sorption Capacities of LDO at Different Temperatures at 100 KPa

temperature (°C)	combined (mmol/g)	reversible (mmol/g)	irreversible (mmol/g)	reversible (%)
100	0.231	0.192	0.039	87.0
200	0.486	0.429	0.057	88.3
300	0.249	0.208	0.041	83.5
400	0.169	0.132	0.037	78.1

chemical interactions.⁴ It is assumed that, at 100 °C, it is mostly a surface phenomena and chemical interactions are restricted probably because of higher activation energy requirements. The interaction between the sorbed CO₂ and the basic sites of the LDO seems to be mainly responsible for the higher values of CO₂ sorption at 200 °C. At increasing temperatures, the sorbent molecules are expected to have higher levels of kinetic energy and have a greater chance to desorb from the surface, thus reducing the sorption value. Therefore, the sorption capacities were measured at 200 °C using the samples calcined at 400 °C for the rest of the studies in this work.

Figure 4a depicts the combined or total CO₂ sorption at 200 °C and distinguishes the reversible and irreversible sorption components for a fresh LDO sample. In general, CO₂ sorption largely occurred at very low pressure (e.g., below 5 kPa) and then gradually increased with the CO₂ pressure. The irreversible sorption contribution was almost constant at 0.05–0.06 mmol/g. The reversible portion, however, increased from 0.19 to 0.43 mmol/g (Figure 4a) in the measured pressure range. This indicates the fact that a major portion of the combined sorption is a relatively weak sorption, making LDOs potentially suitable for pressure swing adsorption. Figure 4b demonstrates the reproducibility of the reversible sorption measurements and the regeneration capability of the material. Reversible sorption measurements were conducted for three times (Ev1, Ev2, Ev3) by repeating vacuum evacuation and CO₂ sorption experiments each time. In all three cases, the reversible sorption values remained at ~0.43 mmol/g, which is 88% of the combined sorption value (0.49 mmol/g) shown as “Fresh” in the figure. The sample was finally regenerated at 400 °C, and the CO₂ sorption was measured. After regeneration, the sorption capacity reverted to 0.48 mmol/g, reaching almost 98% of the value of

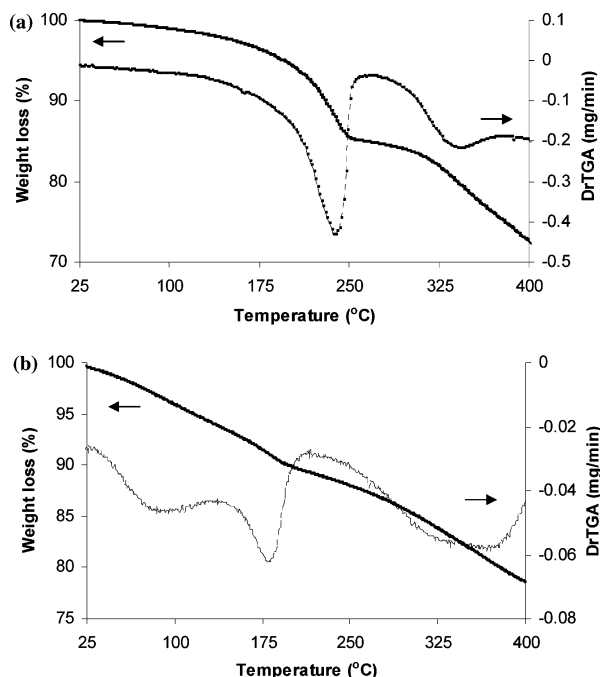


Figure 5. Thermogravimetric analysis showing weight loss and trends of weight loss (DrTGA) for (a) raw LDH and (b) aged LDO.

the freshly calcined sample. This clearly demonstrates the robustness and regenerability of these compounds, which is a highly desirable property, especially for low-cost CO₂-separation applications.

Interestingly, an LDO sample used for CO₂ sorption measurements after it was stored for 48 h under atmospheric conditions showed a significant reduction in the sorption capacity. This prompted us to systematically investigate the negative impacts of postcalcination aging/storage in the atmosphere on the sorption capacity and suggest alternative methods such as vacuum storage and in situ calcination. Comparison of the sorption values of the fresh and aged samples at 200 °C revealed that the aged sample had a systematically lower CO₂ sorption than the fresh LDO sample. However, upon regeneration, the aged LDO sample regained the original CO₂ sorption capacity. For example, the fresh LDO sample had a combined sorption capacity of 0.49 mmol/g, whereas the aged sample had a value at 0.36 mmol/g at 100 kPa. Therefore, it is clear that the sorption patterns are largely affected by the evolution of the structure in the LDO during aging. In fact, the sorbent structure does change to a considerable extent during storage, as discussed in the following section.

Postcalcination Structure Evolution. A systematic study was made to examine the structural changes using characterization techniques such as XRD, FTIR spectroscopy, TGA. Trends of weight loss (DrTGA) reveal that the weight loss patterns for the raw LDH and aged LDO samples are distinctively different, as shown in Figure 5. The amount of weight loss and the decomposed species in different temperature regimes are summarized in Table 2. It is well-known that the weight loss of raw LDH below 190 °C (4.63%) is mainly due to the removal of crystalline water in the interlayer. However, this value is almost doubled (9.68%) in the case of the aged sample. The DrTGA curve in Figure 5b indicates two broad peaks at around 100 and 180 °C, attributed to physically adsorbed water on the surface and crystalline water in the bulk, respectively. This clearly indicates that the LDO adsorbed a significant amount of water from the atmosphere. The weight loss within the temperature range of 190–280 °C, which is primarily attributed

Table 2. Comparison of Percentage Weight Loss of the Species for the Three Samples

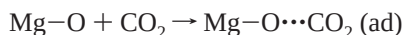
temperature range (°C)	species lost	weight loss (%)		
		raw LDH	aged LDO	fresh LDO
≤ 190	loosely held H ₂ O	4.63	9.68	0.27
190–280	–OH attached to Al and Mg	10.87	3.74	0.57
280–400	–OH attached to Mg and CO ₃ ^{2–}	11	8.05	0.59
total		26.5	21.47	1.44

to the removal of –OH groups from Al–OH–Mg network is 10.87% for the raw sample and 3.74% for the aged sample. This suggests that there is a lower Al–OH–Mg network in the aged sample than in the raw sample. Furthermore, the weight loss in the temperature range of 280–400 °C, which accounts for the decomposition of –OH groups in Mg–OH–Mg and of –CO₃ groups, is 11.0% for the raw sample and 8.05% for the aged sample. This also indicates that there is a considerable amount of Mg–OH in addition to some carbonate in the aged sample. The presence of a high percentage of water, Mg–OH, and a partial Al–OH–Mg network might be responsible for the decrease in the overall CO₂ sorption capacity.^{4,18–21}

The Mg–Al LDH normally undergoes dehydration, dehydroxylation, and decarbonation in series/parallel, leading to the LDO structure during calcination. The LDO is liable to regenerate the LDH structure at room temperature in contact with the atmosphere or water. The regeneration rate from the LDO phase to the LDH phase decreases with the temperature of calcination. Exposure to the atmosphere for at least 17 h or direct contact with liquid water is necessary for the regeneration of Mg–Al oxide derived at 300–500 °C.^{19,31} This is clearly similar to the case of our aged sample that was exposed to the atmosphere for 48 h to partially restore the LDH phase. The FTIR spectra and XRD patterns of aged LDO samples shown in Figure 2 clearly indicate a considerable change in composition and chemical structure when compared to the fresh LDO. The FTIR spectrum for the aged sample is comparable to that of the raw LDH even though it is not an exact replication. In particular, the carbonate band for the aged LDO is similar to that from the raw LDH. These changes indicate the evolution of the chemical structure in the direction of structural revival at least partially during storage of the sample. This is further supported by the XRD patterns for the aged sample shown in Figure 2. As discussed previously, the XRD patterns of the raw LDH show sharp peaks indicating highly crystalline structure, whereas the fresh LDO shows a poorly crystalline MgO phase. Interestingly, the aged LDO shows patterns featured for both LDH and MgO phases. We can see two characteristic peaks of the MgO phase (52° and 75°) and also more featured peaks of the LDH (14°, 28°, 42° etc.). This suggests that there is a partial structural reconstruction from the LDO, which is an amorphous mixed oxide solution, toward the crystalline LDH form upon aging.

Impact of Structure Evolution on CO₂ Sorption. Evolution of the structure taking place during and after the thermal decomposition of the Mg–Al–CO₃ LDH is very important for understanding its CO₂ sorption properties. In the crystalline phase, LDH consists of positively charged Mg–Al–OH brucite-type layers in an octahedral network.¹⁴ The surface and structural properties of LDH in this phase are not suitable for reversible CO₂ sorption, as the Mg–OH-rich network in the crystalline LDH has been found to favor base-catalyzed reactions.¹⁹ Hence, the sorption in this phase is mostly attributed to the acid–base reaction of Mg–OH and CO₂ forming Mg(HCO₃)₂, which is considered to be irreversible sorption.⁴

On the other hand, LDH completely loses interlayer water and dehydroxylates to a large extent upon heating to 400 °C, leading to the formation of a mixed oxide with a three-dimensional network. Although a fair amount of carbonate still remains in the structure, this gives a relatively high surface area and exposes sufficient basic sites on the surface.^{4,19–21} These basic sites favor reversible CO₂ sorption in a form as follows



The interaction between the sorbed CO₂ and the basic sites seems to be mainly responsible for CO₂ sorption at 200 °C. The experimental results obtained in our studies suggest that this interaction is not as weak as in the case of zeolite²⁹ but not as strong as in the case of alkali metal oxide.³⁰ These results strongly suggest the potential of LDO use as a CO₂ sorbent in flue gas systems at temperatures around 200 °C and above.

However, the aged LDO accumulates significant mass during its aging period because of interactions with atmospheric water and CO₂. This led to the hydroxylation of metal oxides, particularly on the surface, with a partial revival of the crystalline LDH structure in addition to the physical bonding of water.^{19,31} The partial reconstruction of the mixed oxide to the LDH phase on the surface of the aged sample thus decreases the CO₂ sorption capacity. As a consequence, recalcination of the aged LDO restores both the amorphous mixed oxide and almost 98% of its original CO₂ sorption capacity. On account of these observations, either vacuum storage of the LDO or in situ calcination methods were used for all further CO₂ sorption measurements. In the in situ calcination method, LDH was calcined in conjunction with CO₂ sorption measurements. These methods produced highly consistent sorption values.

Conclusions

Progressive evolution of the chemical structure from a crystalline LDH phase to an amorphous mixed oxide (LDO) phase was observed during heat treatment. Three-dimensional mixed oxide derivatives of LDH exhibit desirable structural and surface properties for reversible CO₂ sorption. A calcination temperature of 400 °C and a sorption temperature of 200 °C were observed to be the optimal conditions for obtaining the maximum CO₂ sorption capacity. The reversible and irreversible sorption contributions determined demonstrate that about 88% of the combined sorption is reversible sorption at 100 kPa. Aging of the LDO for up to 48 h under normal atmospheric conditions produced significant changes in the structure, resulting in a 25% reduction in the combined sorption capacity. Regeneration of the aged LDO samples restored 98% of the original sorption capacity. Therefore, in situ calcination and simultaneous CO₂ sorption make the system more efficient than precalcination and storage of the samples. Regeneration capability is another important feature of these materials in high-temperature CO₂-separation applications.

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